

IRAI-WP-007

The Legitimacy of Runtime Authority

A Trust Model for Independent Runtime Authority Infrastructure

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Executive Summary

Independent Runtime Authority Infrastructure (IRAI) introduces a new institutional responsibility: determining whether delegated authority remains legitimately exercisable immediately before an autonomous action becomes an institutional consequence.

As Runtime Authority emerges as a distinct architectural discipline, an immediate question follows:

Why should the Runtime Authority determination itself be trusted?

This paper argues that legitimacy does not arise from independence alone. Independence removes conflicts of interest but does not establish correctness.

Instead, Runtime Authority derives legitimacy from deterministic reconstruction. Every determination must be reconstructable from authoritative evidence available at the execution bind point, independently reproducible by third parties, bounded by institutional mandate, and fully evidentially verifiable.

The paper presents the first formal trust model for Independent Runtime Authority Infrastructure and defines the constitutional principles under which Runtime Authority determinations may legitimately admit, escalate or refuse execution.

1. Introduction

Earlier papers established three foundational propositions.

First, governance decisions made during planning, approval and delegation cannot guarantee that authority remains exercisable when execution eventually occurs.

Second, an independent Runtime Authority layer is required to determine admissibility immediately before institutional consequence forms.

Third, the FlowSignal MVP demonstrates that such determinations can be implemented deterministically within an operational system.

The successful implementation of Runtime Authority naturally raises a deeper architectural question.

Why should the Runtime Authority itself be trusted?

This question has emerged consistently across independent reviews from enterprise governance specialists, healthcare leaders, AI safety researchers, ethicists and systems architects. While the terminology differs, the underlying concern is remarkably consistent: if Runtime Authority becomes the institutional decision immediately preceding consequence formation, what establishes the legitimacy of that determination?

This paper addresses that question.

Rather than proposing trust through institutional status or organisational authority, it argues that Runtime Authority derives legitimacy through deterministic reconstruction of institutional admissibility using authoritative evidence available at the execution bind point.

2. The Architectural Problem

The emergence of Independent Runtime Authority Infrastructure introduces a new institutional responsibility. Unlike governance systems, which establish policies and delegated authority, Runtime Authority determines whether that delegated authority remains legitimately exercisable immediately before execution.

As soon as Runtime Authority assumes this responsibility, an immediate architectural question follows:

What makes the Runtime Authority determination itself legitimate?

The intuitive answer is often independence.

Runtime Authority is deliberately separated from business workflows, application logic and autonomous decision-making. This architectural independence prevents conflicts of interest and allows authority to be evaluated without influence from the system requesting execution.

Independence, however, is only a necessary condition.

It is not a sufficient one.

An independent system can still produce incorrect, incomplete or unjustified determinations. Architectural separation alone provides no assurance that the determination accurately reflects the institution's legitimate authority at the execution bind point.

The institutional problem is therefore considerably deeper.

Runtime Authority cannot simply assert that execution should proceed.

It must demonstrate why that determination is institutionally valid.

This distinction separates Runtime Authority from traditional enforcement mechanisms.

Many existing control systems determine whether execution complies with predefined policy. Runtime Authority addresses a different question. It determines whether the institution itself still possesses legitimate authority to allow the requested consequence to occur under the current operational conditions.

That determination cannot rely upon historical approval alone.

Nor can it rely solely upon workflow state, system ownership or policy configuration.

Instead, legitimacy must be reconstructed from the authoritative evidence that exists at the precise moment execution is requested.

Only then can the determination be considered institutionally defensible.

This leads to a fundamental principle of Independent Runtime Authority Infrastructure:

Runtime Authority does not create authority. It reconstructs whether legitimate authority still exists at the execution bind point.

Everything that follows in this paper builds upon that principle.

The legitimacy of Runtime Authority therefore derives neither from its independence nor from institutional assertion. It derives from the deterministic reconstruction of admissibility using authoritative evidence that can be independently verified, reproduced and challenged by others.

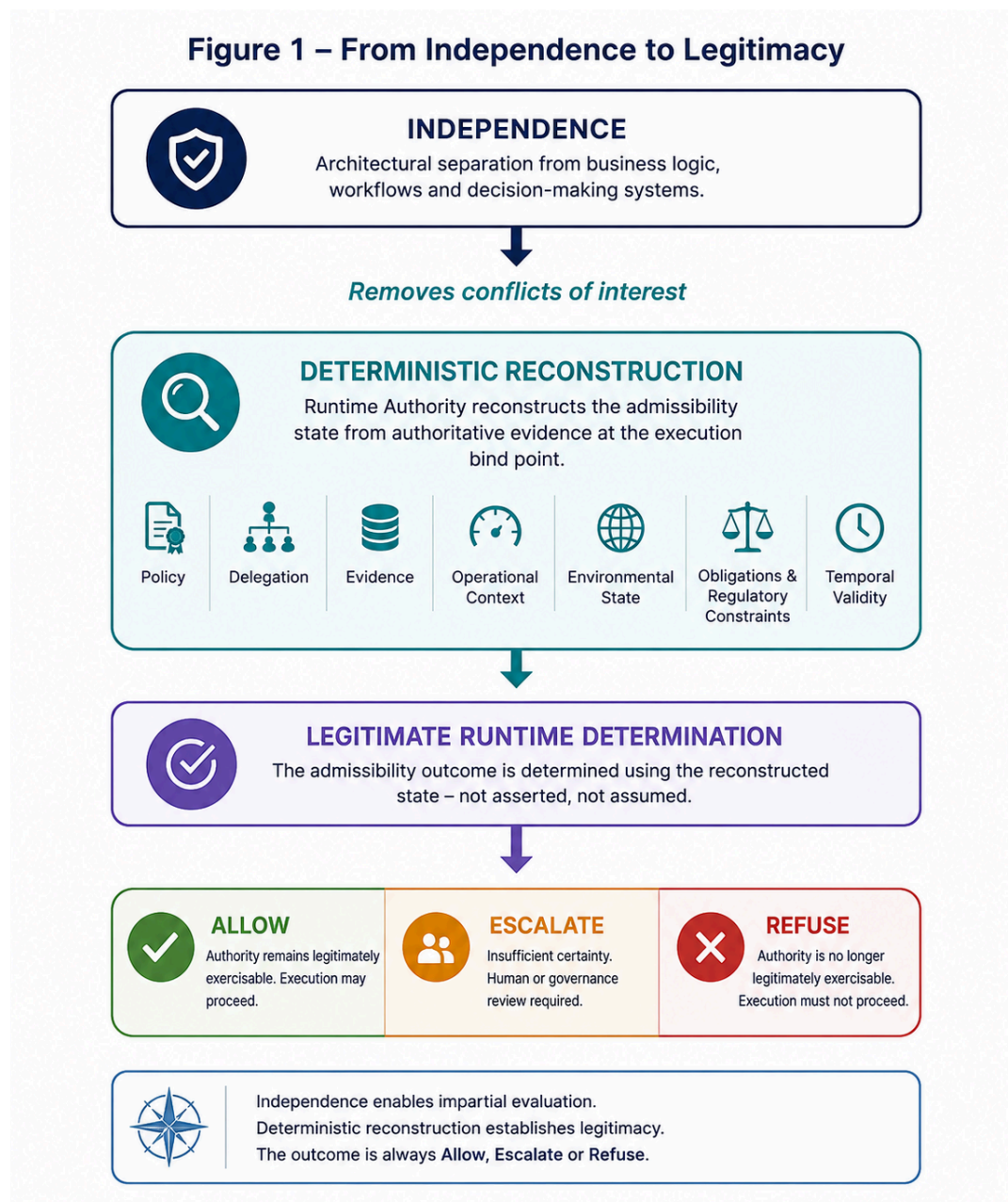


Figure 1. From Independence to Legitimacy

This figure illustrates the architectural progression by which Runtime Authority transforms independent evaluation into a legitimate execution decision. Architectural independence alone does not establish legitimacy; rather, it enables an impartial determination free from application or workflow bias. At the execution bind point, Runtime Authority deterministically reconstructs the admissibility state using authoritative evidence, including policy, delegation, operational context, environmental conditions, regulatory obligations, and temporal validity. Based on this reconstructed state, the system produces one of three deterministic outcomes—**ALLOW**, **ESCALATE**, or **REFUSE**—representing whether delegated authority remains legitimately exercisable immediately before consequence formation. The figure emphasises that legitimacy is not asserted or assumed, but reconstructed from evidence at runtime.

Constitutional Standing

Demonstrating institutional legitimacy first requires Runtime Authority to possess legitimate standing to make the determination itself.

Runtime Authority does not possess standing because it is trusted. It possesses standing because governance delegates to it the bounded constitutional responsibility of reconstructing institutional admissibility immediately before consequence formation.

Its standing is therefore derivative rather than original. Runtime Authority does not originate institutional authority, nor does it redefine governance intent. It exercises only the narrowly defined responsibility delegated to it by the institution: to determine whether previously delegated authority remains legitimately exercisable at the execution bind point.

This distinction is fundamental. Governance remains constitutionally responsible for defining authority, policy and obligations. Runtime Authority is constitutionally responsible only for evaluating whether those conditions continue to be satisfied immediately before consequence formation.

Runtime Authority cannot determine legitimacy merely because it is independent. It can determine legitimacy because the institution has explicitly delegated to it the bounded constitutional responsibility of reconstructing institutional admissibility immediately before consequence formation. Independence does not create that authority; it preserves the integrity of its exercise by ensuring that the determination is made free from operational, commercial and application influence.

Only once this constitutional standing has been established can the legitimacy of an individual Runtime Authority determination be assessed.

3. Runtime Legitimacy Requires Reconstruction, Not Assertion

Architectural independence establishes the conditions under which a Runtime Authority determination can be trusted, but independence alone does not make that determination legitimate.

The central architectural question therefore becomes:

How does an independent Runtime Authority establish that delegated authority remains legitimately exercisable at the exact moment execution is about to occur?

The answer is neither policy evaluation nor workflow orchestration.

Instead, Runtime Authority performs a deterministic reconstruction of the execution conditions immediately before institutional consequence forms.

Unlike traditional governance systems, which primarily evaluate whether an action was approved, Runtime Authority evaluates whether every condition required for that approval to remain executable still exists.

This distinction is fundamental.

Approval represents a historical governance decision.

Runtime Authority evaluates the present institutional reality.

Those two states frequently diverge.

For example:

- Delegation may have expired.
- Regulatory obligations may have changed.
- New evidence may invalidate previous assumptions.
- Environmental conditions may have altered operational risk.
- A supervising clinician or authorised officer may no longer be available.
- The institutional mandate itself may have been withdrawn.

Any one of these changes may leave the original approval intact while simultaneously making execution institutionally inadmissible.

Runtime Authority therefore reconstructs what may be described as the **Institutional Admissibility State**.

Rather than relying upon previously stored decisions, it deterministically evaluates whether the complete set of admissibility conditions continues to exist immediately before execution.

This reconstruction typically considers multiple categories of institutional evidence, including:

- Delegated authority and mandate validity.
- Applicable organisational policy.
- Regulatory and legal obligations.
- Operational context.
- Environmental conditions.
- Temporal validity.
- Required human supervision or oversight.
- Availability and integrity of supporting evidence.

The precise admissibility criteria remain implementation-specific.

Healthcare, financial services, government and critical infrastructure each evaluate different institutional conditions.

However, the architectural responsibility remains identical:

To reconstruct sufficient evidence to determine whether authority continues to exist at execution time.

Only after this reconstruction can Runtime Authority deterministically produce one of three admissibility outcomes:

- **ALLOW** — delegated authority remains legitimately exercisable.
- **ESCALATE** — insufficient certainty exists to determine admissibility.
- **REFUSE** — delegated authority no longer exists for this execution.

The legitimacy of the decision therefore derives not from independence itself, but from the evidence-based reconstruction performed immediately before consequence formation.

This distinction explains why Runtime Authority should not be viewed as another policy engine.

Policy contributes evidence.

Runtime Authority determines whether that evidence is collectively sufficient for institutional consequence to proceed.

Key Principle

Runtime Authority does not assert legitimacy. It reconstructs legitimacy from institutional evidence immediately before execution.

4. The Architectural Responsibility of Runtime Authority

One of the recurring themes emerging across healthcare, enterprise governance, AI safety and institutional architecture is that Runtime Authority is frequently misunderstood as another governance engine.

It is not.

Its responsibility is considerably narrower.

Equally, its responsibility is considerably more precise.

Runtime Authority exists to answer a single institutional question:

Does legitimate authority still exist for this consequence to be allowed to form?

Everything else belongs elsewhere.

This distinction is important because modern governance architectures already contain numerous specialised components responsible for policy management, compliance, workflow orchestration, identity management, audit, monitoring and decision support.

Runtime Authority neither replaces nor duplicates these functions.

Instead, it depends upon them.

Policies define institutional expectations.

Identity systems establish who holds delegated authority.

Workflow engines coordinate execution.

Monitoring platforms observe system behaviour.

Evidence repositories preserve operational records.

Risk systems identify changing conditions.

Each contributes information describing the institutional state.

Runtime Authority consumes that information without assuming ownership of it.

Its architectural responsibility begins only when execution is imminent.

At the execution bind point, Runtime Authority reconstructs the admissibility state using evidence supplied by these surrounding systems.

It does not rewrite policy.

It does not allocate authority.

It does not perform workflow orchestration.

It does not calculate business outcomes.

It does not optimise execution.

Its responsibility is limited to determining whether the institutional conditions required for legitimate execution continue to exist.

This separation is what preserves architectural independence.

Were Runtime Authority permitted to author policy, modify workflows, assign authority or generate evidence, it would become both judge and participant within the same governance process.

Its independence would therefore become compromised.

Instead, Runtime Authority occupies a deliberately constrained position within the execution architecture.

It reconstructs institutional legitimacy from evidence generated elsewhere.

It evaluates admissibility using deterministic rules.

It produces one of three admissible outcomes:

- **ALLOW** — execution remains institutionally legitimate.
- **ESCALATE** — admissibility cannot be determined with sufficient certainty.
- **REFUSE** — legitimate authority no longer exists.

Once that determination has been produced, responsibility transfers back to the execution platform.

Runtime Authority neither executes nor prevents execution directly.

It determines whether execution remains institutionally admissible.

Execution systems remain responsible for carrying out—or refusing—the resulting action.

The architectural responsibility therefore becomes remarkably simple.

Everything before Runtime Authority contributes evidence.

Runtime Authority reconstructs the institutional conditions under which legitimacy exists.

Everything after Runtime Authority executes the institutional consequence.

This clear separation explains why Runtime Authority complements existing governance architectures rather than competing with them.

It introduces a missing responsibility, not a replacement **discipline**.

Key Principle

Runtime Authority owns only one architectural responsibility: determining whether delegated authority remains legitimately exercisable immediately before institutional consequence forms. Every other governance responsibility remains where it already belongs.

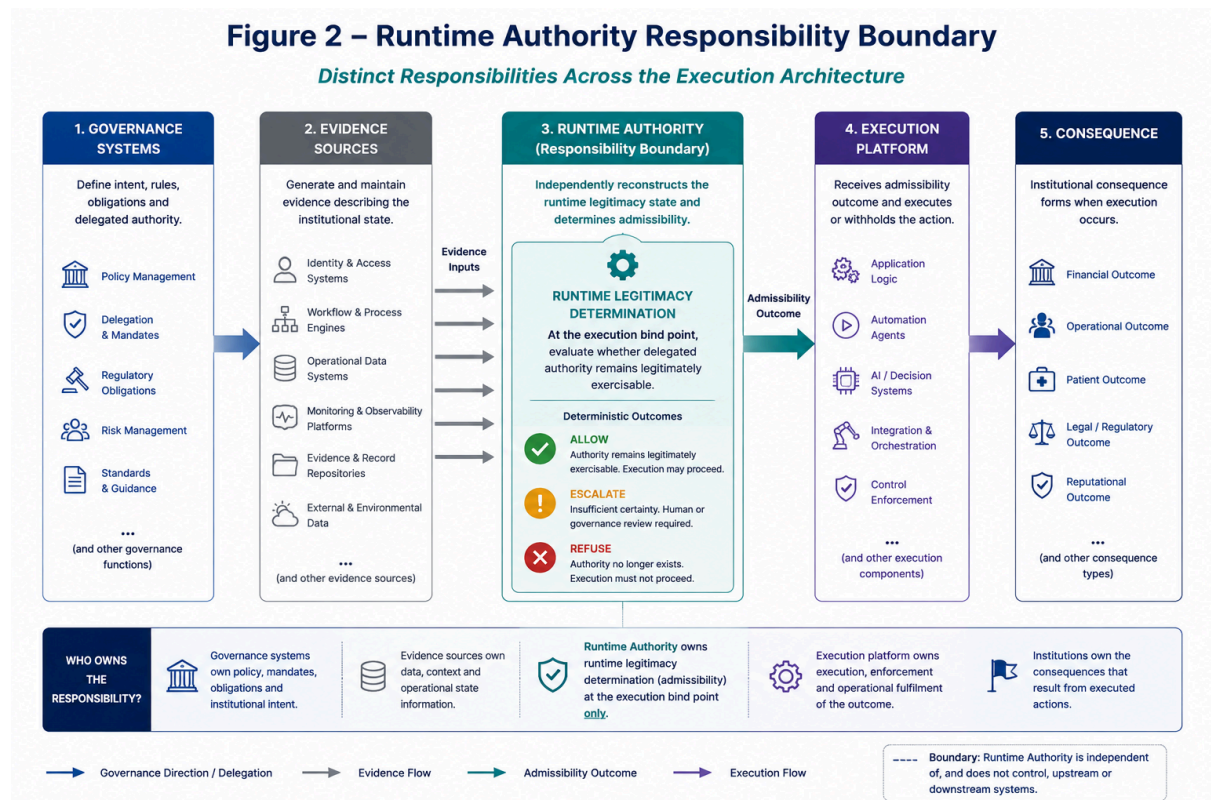


Figure 2. Runtime Authority Responsibility Boundary

This figure illustrates the distinct architectural responsibilities that surround Runtime Authority within an execution architecture. Governance systems establish institutional intent through policy, delegated authority, regulatory obligations and organisational controls, while evidence sources maintain the operational state required to evaluate those intentions. Runtime Authority occupies a deliberately independent position between evidence and execution. At the execution bind point, it reconstructs the Institutional Admissibility State from institutional evidence and deterministically produces one of three admissibility outcomes—**ALLOW**, **ESCALATE**, or **REFUSE**. Responsibility then transfers to the execution platform, which implements the outcome but does not determine its legitimacy. The resulting institutional consequence is owned by the organisation rather than the Runtime Authority itself. The figure emphasises that Runtime Authority introduces a new architectural responsibility—runtime legitimacy determination—without replacing governance, execution, or audit systems, thereby preserving clear separation of concerns and architectural independence.

5. The Institutional Admissibility State.

The legitimacy of an execution decision cannot be stored.

It cannot be assumed.

Nor can it be inherited from an earlier approval.

Instead, legitimacy exists only as the current institutional state immediately before consequence formation.

This state may be described as the **Institutional Admissibility State**.

The Institutional Admissibility State represents the complete set of institutional conditions that must simultaneously exist for delegated authority to remain legitimately exercisable at execution time.

Unlike approval, which records that authority once existed, the Institutional Admissibility State establishes whether authority continues to exist now.

This distinction is subtle but architecturally profound.

Approval is historical.

Legitimacy is contemporary.

Approval is static.

Legitimacy is dynamic.

Approval records institutional intent.

Legitimacy reflects institutional reality.

Consequently, Runtime Authority does not search for a single permission or policy.

Instead, it reconstructs the Institutional Admissibility State from multiple independent sources of institutional evidence.

Those sources may include:

- Delegated authority.
- Applicable policy.
- Regulatory obligations.
- Operational context.
- Environmental conditions.
- Temporal validity.
- Human availability.

- Required evidence.
- Organisational mandates.
- Jurisdictional constraints.
- External dependencies.

The exact composition varies between institutions.

A hospital evaluates different evidence from a bank.

A government agency evaluates different evidence from a manufacturing organisation.

Yet despite these domain differences, the architectural requirement remains identical.

Every institution possesses a Institutional Admissibility State.

Only the evidence required to reconstruct it changes.

Importantly, Runtime Authority does not define what legitimacy should be.

That responsibility belongs to institutional governance.

Governance specifies the admissibility model.

Runtime Authority evaluates whether the institution's own model is currently satisfied.

This separation preserves both institutional autonomy and architectural independence.

Runtime Authority therefore functions as a deterministic evaluator of institutional legitimacy rather than a creator of governance policy.

The Institutional Admissibility State should therefore be understood as an architectural object rather than a business object.

Applications rarely model it.

Workflow engines do not own it.

Policy engines define aspects of it.

Audit systems record evidence about it.

Runtime Authority is responsible for reconstructing it immediately before execution.

Only after this reconstruction can admissibility be determined.

This explains why legitimacy cannot simply be inferred from previous approvals.

The Institutional Admissibility State is continually evolving.

Every execution decision represents a fresh institutional determination.

Key Principle

Approval answers whether authority once existed. The Institutional Admissibility State answers whether authority still exists immediately before consequence formation.

6. Runtime Determinations Must Be Reconstructable

An institutional determination that cannot later be reconstructed cannot be independently verified.

Equally, a determination that cannot be independently verified cannot provide defensible authority for institutional consequence.

This requirement extends beyond auditability.

Traditional audit records explain what occurred.

Runtime Authority must demonstrate why the determination was legitimate at the precise moment execution was permitted or refused.

This distinction is critical.

An audit trail may record that execution occurred.

A Runtime Authority determination must demonstrate that execution remained institutionally admissible immediately before consequence formation.

These are fundamentally different objectives.

Consequently, every Runtime Authority determination should be capable of independent reconstruction using the evidence that existed at the execution bind point.

This does not require replaying an application or recreating an entire workflow.

Instead, it requires reconstructing the Institutional Admissibility State that existed when the admissibility determination was made.

If an independent reviewer receives the same institutional evidence that was available at execution time, the reviewer should be capable of arriving at the same admissibility outcome.

This principle mirrors long-established engineering disciplines.

A financial transaction can be independently reconciled.

A medical diagnosis can be independently reviewed.

A legal judgement can be independently examined against the evidence presented.

Runtime Authority introduces the same expectation for institutional execution.

The admissibility determination should not depend upon hidden application state, undocumented assumptions or opaque reasoning.

It should depend only upon institutional evidence and deterministic evaluation.

This requirement has significant architectural consequences.

Runtime Authority should preserve sufficient evidence to demonstrate:

- the delegated authority that existed;
- the policies and obligations evaluated;
- the operational context at execution time;
- the evidence used to reconstruct the Institutional Admissibility State;
- the deterministic rules applied;
- the resulting admissibility outcome;
- the time at which the determination occurred.

Collectively, these elements enable independent reconstruction of the admissibility decision.

Importantly, reconstruction is not the same as explanation.

An explanation may describe why a decision was made.

Reconstruction demonstrates that the decision was the only admissible outcome given the institutional state that existed at that moment.

This distinction elevates Runtime Authority from a governance component to an evidentiary component.

Its purpose is not merely to determine legitimacy, but to enable that determination to withstand later institutional scrutiny.

Whether the review is conducted by an internal governance committee, a regulator, an auditor or a court, the admissibility determination should remain reproducible from the preserved evidence.

The institution is therefore not asked to trust the Runtime Authority.

It is asked to verify it.

Key Principle

A Runtime Authority determination is trustworthy not because it is asserted, but because it can be independently reconstructed from the institutional evidence that existed immediately before consequence formation.

7. The Authority Receipt

A Runtime Authority determination is transient.

An institutional consequence is permanent.

For this reason, every admissibility determination should produce a durable evidentiary artefact describing the institutional state that existed immediately before execution.

This artefact may be referred to as an Authority Receipt.

The Authority Receipt is not an audit log.

Nor is it simply a decision record.

Instead, it is the evidentiary package that allows an independent party to reconstruct why execution was admitted, escalated or refused.

Its purpose is not to justify a decision retrospectively.

Its purpose is to preserve the institutional evidence required to independently verify that the determination was legitimate when it was made.

This distinction is significant.

Traditional logging records what happened.

An Authority Receipt records why execution remained institutionally admissible at the execution bind point.

Consequently, an Authority Receipt should capture the evidence required to reconstruct the Institutional Admissibility State.

While implementations will differ, a complete Authority Receipt will typically preserve:

- the unique execution identifier;
- the delegated authority under evaluation;
- the applicable policy and institutional obligations;
- the operational context at the execution bind point;
- references to supporting evidence;
- the Institutional Admissibility State reconstructed by the Runtime Authority;
- the deterministic admissibility outcome (ALLOW, ESCALATE, or REFUSE);
- the execution timestamp;
- the Runtime Authority version responsible for the determination;
- cryptographic integrity information sufficient to detect subsequent modification.

Importantly, the Authority Receipt does not seek to preserve every operational detail.

Its objective is considerably narrower.

It preserves only the evidence necessary to independently reconstruct the admissibility determination.

This enables a regulator, auditor, investigator or institutional reviewer to ask a single question:

Given the evidence that existed at this moment, was this execution legitimately admissible?

If the preserved evidence produces the same deterministic outcome, institutional confidence in the Runtime Authority is strengthened.

If it does not, the determination can be challenged using objective evidence rather than opinion.

This capability becomes increasingly important as AI systems begin making decisions whose consequences extend beyond individual applications.

Healthcare, financial services, government and critical infrastructure all require evidence that survives organisational change, software evolution and regulatory scrutiny.

The Authority Receipt provides this continuity.

It becomes the portable evidentiary record linking governance intent, runtime legitimacy determination and institutional consequence.

In this way, Runtime Authority does not merely decide.

It leaves behind sufficient evidence for that decision to be independently understood, verified and defended long after execution has completed.

Key Principle

The Authority Receipt preserves not merely the outcome of a Runtime Authority determination, but the institutional evidence required to independently reconstruct why that outcome was legitimate at the moment execution occurred.

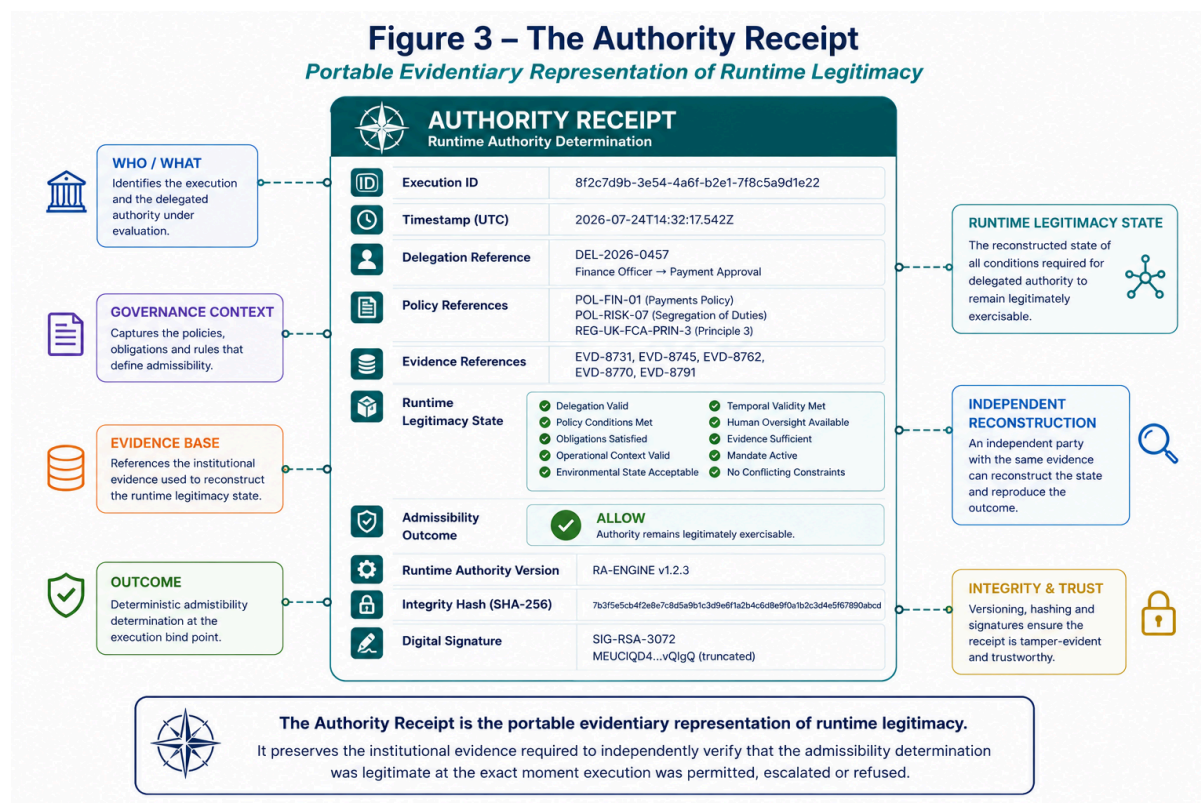


Figure 3. The Authority Receipt

This figure illustrates the structure and purpose of the **Authority Receipt**, the portable evidentiary artefact produced following every Runtime Authority determination. Rather than acting as an audit log or execution record, the Authority Receipt preserves the institutional evidence required to independently reconstruct the Institutional Admissibility State at the execution bind point. It captures the identity of the delegated authority under evaluation, the applicable governance policies, evidence references, reconstructed legitimacy conditions, deterministic admissibility outcome (**ALLOW**, **ESCALATE**, or **REFUSE**), execution timestamp, Runtime Authority version, and cryptographic integrity information. Together, these elements enable regulators, auditors, investigators, or institutional reviewers to independently verify that the admissibility determination was legitimate based solely on the evidence available immediately before consequence formation. The Authority Receipt therefore provides the enduring evidentiary link between governance intent, runtime legitimacy determination, and institutional consequence, ensuring that Runtime Authority decisions remain transparent, reproducible, and defensible long after execution has occurred.

8. Independent Verification and Institutional Trust

The existence of an Authority Receipt does not, by itself, establish institutional trust.

It merely preserves the evidence required to examine a Runtime Authority determination.

Institutional trust arises only when that preserved evidence enables an independent party to verify that the admissibility determination was correct.

This distinction is fundamental.

Reconstruction answers:

Can the institutional conditions that existed at execution time be reproduced?

Verification answers:

Does that reconstructed state justify the admissibility outcome that was reached?

These are related but distinct architectural activities.

A Runtime Authority implementation therefore should not expect its determinations to be accepted on assertion alone.

Instead, each determination should be capable of independent verification by parties external to the Runtime Authority itself.

Those parties may include:

- Internal governance teams.
- Clinical safety officers.
- Enterprise risk functions.
- Independent auditors.
- Regulatory authorities.
- Judicial reviewers.
- External assurance providers.

Each begins with the same evidence.

Each reconstructs the same Institutional Admissibility State.

Each should arrive at the same admissibility outcome.

This repeatability is what transforms Runtime Authority from an implementation into institutional infrastructure.

Importantly, independent verification does not require an identical software implementation.

Different Runtime Authority engines may evaluate the same admissibility model provided they reconstruct the same institutional evidence and apply equivalent deterministic rules.

This is analogous to financial reconciliation.

Multiple accounting systems may independently verify the same transaction without sharing identical software.

Likewise, Runtime Authority should support institutional interoperability through shared evidence rather than shared implementation.

This distinction has significant architectural implications.

Institutions remain free to define their own governance models.

They remain free to implement their own Runtime Authority platforms.

What becomes portable is the evidence.

Consequently, institutional trust shifts away from software ownership and towards evidentiary integrity.

The Authority Receipt becomes the common object through which independent verification occurs.

Rather than asking:

Do we trust this Runtime Authority?

Institutions instead ask:

Can we independently verify the admissibility determination from the preserved evidence?

This represents an important change in how trust is established.

Trust no longer depends upon confidence in a vendor, application or platform.

Trust depends upon whether the institutional evidence deterministically supports the outcome that was produced.

The architectural consequence is profound.

Runtime Authority is no longer simply making decisions.

It is producing determinations that remain independently verifiable across organisational, regulatory and technological boundaries.

This capability becomes increasingly important as autonomous systems begin interacting across institutional ecosystems rather than remaining confined within a single organisation.

Key Principle

Institutional trust does not arise because Runtime Authority is independent. It arises because independent parties can verify that the preserved institutional evidence deterministically supports the admissibility outcome.

9. Towards an Open Runtime Authority Interface (ORAI)

If Runtime Authority is to become a recognised architectural responsibility, institutions must avoid creating isolated implementations that cannot interoperate.

History demonstrates that architectural disciplines mature through standardised interfaces rather than standardised products.

The Internet did not emerge because everyone deployed identical routers.

Identity federation did not emerge because organisations standardised upon a single identity provider.

Financial messaging did not require every bank to implement identical software.

Instead, interoperability emerged through common interfaces that allowed independent implementations to exchange trusted information.

Runtime Authority should evolve in the same manner.

The architectural responsibility is independent.

Its implementation should remain independent.

However, the evidence describing its determinations should become portable.

This observation motivates the concept of an Open Runtime Authority Interface (ORAI).

ORAI is not a Runtime Authority implementation.

Nor is it another governance platform.

Instead, ORAI represents a vendor-neutral interoperability contract describing how Runtime Authority determinations can be exchanged, verified and consumed across heterogeneous execution environments.

Its purpose is deliberately narrow.

It enables independently developed Runtime Authority systems to communicate admissibility determinations without requiring common software implementations.

Rather than prescribing algorithms or governance models, ORAI would define the structure of the evidentiary objects exchanged between systems.

Those objects may include:

- Authority Receipts.

- Institutional Admissibility State references.
- Admissibility outcomes.
- Evidence references.
- Integrity metadata.
- Verification information.
- Version identifiers.

Each institution remains responsible for defining its own governance policies, admissibility models and Runtime Authority implementation.

ORAI does not standardise institutional judgement.

It standardises how institutional determinations are represented and exchanged.

This distinction preserves institutional sovereignty while enabling interoperability.

For example, a healthcare provider, financial institution and government department may each operate different Runtime Authority implementations based upon different regulatory obligations.

Yet all may exchange Authority Receipts using a common evidentiary representation.

Each institution independently determines legitimacy.

Each institution independently verifies legitimacy.

ORAI simply enables those determinations to be understood across organisational boundaries.

This approach also supports future ecosystem development.

Execution platforms, workflow engines, governance systems and audit services need not embed proprietary Runtime Authority logic.

Instead, they consume deterministic admissibility determinations through a stable architectural interface.

The result is an ecosystem in which Runtime Authority implementations compete on quality, assurance and performance while remaining interoperable through shared evidentiary standards.

Such an approach mirrors the evolution of many mature technology disciplines.

Innovation occurs within implementations.

Interoperability occurs through open contracts.

Key Principle

ORAI does not standardise Runtime Authority implementations. It standardises the evidentiary interface through which independently implemented Runtime Authority systems exchange and verify admissibility determinations.

10. Constitutional Principles of Runtime Authority

The legitimacy of Runtime Authority does not arise solely from its implementation.

Instead, it derives from adherence to a set of constitutional principles that define both the limits and responsibilities of the architectural discipline.

These principles are intentionally implementation-independent.

They do not prescribe software design, programming languages or deployment models.

Rather, they establish the institutional constraints under which any Runtime Authority implementation may legitimately determine admissibility immediately before consequence formation.

Collectively, these principles preserve the independence, determinism and institutional legitimacy of Runtime Authority.

Principle 1 — Runtime Authority Shall Not Create Authority

Runtime Authority is not a source of institutional authority.

It cannot allocate, delegate or invent authority where none exists.

Its responsibility is limited to determining whether delegated authority remains legitimately exercisable using the institutional evidence available at the execution bind point.

Authority originates within institutional governance.

Runtime Authority merely evaluates whether that authority continues to exist.

Principle 2 — Runtime Authority Shall Not Create Evidence

Runtime Authority does not manufacture operational evidence.

Evidence originates from authoritative institutional systems including governance platforms, identity services, workflow engines, operational systems and trusted external sources.

Runtime Authority consumes this evidence.

It never becomes its owner.

This separation preserves evidentiary independence and prevents Runtime Authority from validating evidence that it has itself produced.

Principle 3 — Runtime Authority Shall Not Reinterpret Governance

Institutional governance determines organisational intent.

Policies define admissibility.

Delegation establishes authority.

Regulation establishes obligation.

Runtime Authority does not reinterpret these institutional decisions.

It deterministically evaluates whether the institution's own admissibility model remains satisfied at execution time.

Interpretation belongs to governance.

Evaluation belongs to Runtime Authority.

Principle 4 — Runtime Authority Shall Not Compensate for Missing Evidence

Where sufficient evidence does not exist to reconstruct the Institutional Admissibility State, Runtime Authority must not infer missing information.

Institutional ambiguity must never be resolved through assumption.

Instead, insufficient evidence constitutes a deterministic admissibility outcome.

Where legitimacy cannot be established with sufficient certainty, Runtime Authority shall produce an ESCALATE determination.

The absence of evidence is itself institutional evidence.

Principle 5 — Runtime Authority Shall Remain Deterministic

Identical institutional evidence must always produce identical admissibility outcomes.

The Runtime Authority determination must not vary according to operator preference, implementation discretion or execution environment.

Determinism is fundamental to independent verification.

Without determinism, legitimacy cannot be reconstructed.

Principle 6 — Runtime Authority Shall Preserve Independent Verifiability

Every admissibility determination must preserve sufficient evidence to enable independent reconstruction.

The institution should never be required to trust the Runtime Authority solely because it produced the determination.

Instead, an independent reviewer should be able to reproduce the Institutional Admissibility State and verify the admissibility outcome using the preserved Authority Receipt and supporting evidence.

Trust is therefore established through verification rather than assertion.

Principle 7 — Runtime Authority Shall Not Execute Institutional Consequences

Runtime Authority determines admissibility.

It does not perform execution.

Execution remains the responsibility of the execution platform or authorised institutional actor.

This separation preserves the distinction between determining whether execution is legitimate and carrying out the resulting consequence.

Runtime Authority remains an independent institutional decision-maker rather than an execution engine.

Principle 8 — Runtime Authority Shall Preserve Institutional Accountability

The existence of Runtime Authority does not transfer institutional responsibility.

Responsibility for governance remains with governance.

Responsibility for execution remains with the execution platform.

Responsibility for institutional consequence remains with the institution itself.

Runtime Authority contributes an independent admissibility determination.

It does not replace organisational accountability.

Principle 9 — Runtime Authority Shall Remain Implementation Independent

The constitutional principles described in this paper apply irrespective of implementation technology, vendor or deployment model.

Multiple Runtime Authority implementations may legitimately coexist provided each satisfies the constitutional responsibilities defined herein.

This independence allows institutional innovation while preserving a common architectural discipline.

Principle 10 — Runtime Authority Exists to Preserve Institutional Legitimacy

The ultimate purpose of Runtime Authority is not automation.

It is not efficiency.

It is not governance orchestration.

Its purpose is singular:

To determine whether delegated authority remains legitimately exercisable immediately before institutional consequence forms.

Every architectural decision, evidentiary requirement and implementation choice should reinforce this constitutional objective.

Constitutional Summary

These principles collectively define Runtime Authority as a bounded architectural responsibility rather than a discretionary decision-making system.

Runtime Authority does not invent authority, generate governance or replace institutional judgement.

Instead, it reconstructs the Institutional Admissibility State using authoritative evidence, deterministically evaluates admissibility and preserves sufficient evidence to enable independent verification.

Any implementation unable to satisfy these constitutional principles may still perform runtime evaluation, but it should not be regarded as an implementation of Independent Runtime Authority Infrastructure as defined by this paper.

Key Principle

The legitimacy of Runtime Authority derives not from the software that implements it, but from the constitutional principles that constrain how it reconstructs, determines and evidences institutional admissibility.

11. Runtime Authority as an Architectural Discipline

Throughout the evolution of enterprise computing, new architectural responsibilities have emerged whenever existing disciplines proved insufficient to address new operational realities.

Identity management emerged when applications could no longer manage users independently.

Public key infrastructure emerged when digital trust could no longer rely on isolated implementations.

API gateways emerged when services required consistent mediation across distributed systems.

In each case, the architectural responsibility existed before the technology category became widely recognised.

Runtime Authority appears to represent a similar transition.

The rapid adoption of autonomous and AI-assisted systems has exposed an architectural responsibility that traditional governance disciplines implicitly assumed but rarely modelled explicitly.

Governance determines what should be permitted.

Execution determines what will occur.

Between these two responsibilities lies a distinct institutional question:

Does legitimate authority still exist for this consequence to be allowed to form?

This question cannot be fully answered by governance alone.

Nor can it be delegated to applications, workflow engines or AI models whose primary responsibility is execution rather than institutional legitimacy.

Instead, it requires an independent architectural capability whose sole purpose is to reconstruct and evaluate the Institutional Admissibility State immediately before consequence formation.

This paper has argued that Runtime Authority fulfils that responsibility.

Its purpose is intentionally narrow.

It does not replace governance.

It does not orchestrate execution.

It does not generate institutional policy.

It determines whether delegated authority remains legitimately exercisable at the precise moment execution is about to occur.

The concepts introduced throughout this paper—including the Institutional Admissibility State, Authority Receipts, independent verification and implementation-independent interoperability—collectively describe more than a software component.

They describe an architectural discipline.

Like identity management or public key infrastructure before it, Runtime Authority defines a persistent responsibility that remains independent of any individual product or vendor implementation.

Different organisations will inevitably develop different Runtime Authority platforms.

Different industries will define different admissibility models.

Different regulators will establish different evidentiary expectations.

Yet the architectural responsibility remains unchanged.

Where autonomous systems can produce institutional consequences, institutions require an independent means of determining whether execution remains legitimate immediately before those consequences form.

This requirement exists independently of technology stacks, implementation languages, organisational structures or regulatory domains.

It is therefore better understood as a permanent architectural responsibility than as a temporary governance innovation.

Whether the terminology introduced in this paper ultimately becomes widely adopted is less important than the underlying architectural distinction itself.

The increasing convergence of perspectives from healthcare, enterprise architecture, AI governance, safety engineering, law and institutional design suggests that practitioners are independently identifying the same boundary.

The remaining challenge is not to convince institutions that this responsibility exists.

It is to define its boundaries, interfaces and evidentiary standards with sufficient clarity that independent implementations can emerge while remaining interoperable and institutionally trustworthy.

That work has only just begun.

Key Principle

Runtime Authority should be understood not as another governance product, but as an independent architectural discipline responsible for determining whether delegated authority remains legitimately exercisable immediately before institutional consequence forms.

12. Conclusion

Modern governance architectures have become increasingly effective at defining policy, allocating authority and recording accountability.

However, they continue to rely upon an assumption that authority granted during planning or approval remains valid until execution occurs.

For autonomous and AI-assisted systems, that assumption can no longer be safely made.

Authority is not static.

It evolves with operational context, evidence, regulation, delegation and institutional state.

The architectural challenge is therefore no longer simply to explain decisions after they occur.

It is to determine whether execution remains legitimate immediately before institutional consequence forms.

This paper has argued that this responsibility is distinct from governance, workflow orchestration, policy evaluation and audit.

It belongs to an independent Runtime Authority layer whose purpose is to reconstruct the Institutional Admissibility State, deterministically evaluate admissibility and preserve the evidence necessary for independent verification.

By introducing concepts such as the Institutional Admissibility State, the Authority Receipt and the Open Runtime Authority Interface (ORAI), this paper proposes a framework through which Runtime Authority can evolve as an implementation-independent architectural discipline.

The intent is not to prescribe a single implementation or governance model.

Rather, it is to establish a common architectural vocabulary that enables institutions to recognise, design and independently verify this previously implicit responsibility.

As AI systems become increasingly autonomous and institutional consequences become more immediate, the ability to determine **whether execution should still be permitted** may prove as important as the ability to explain **why execution occurred**.

Runtime Authority is not another governance layer.

It is the constitutional mechanism through which institutions determine whether delegated authority remains legitimately exercisable immediately before consequence formation.

Governance creates authority.

Runtime Authority reconstructs whether that authority still exists.

That distinction defines a permanent architectural responsibility for autonomous systems.